EE4-23
EE9-CS2-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Friday, 8 May 10:00 am

Time allowed: 3:00 hours

Corrected Copy

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following second order differential equation of a single real unknown:

$$\ddot{y}(t) - (1 - y^2(t) - \dot{y}^2(t))\dot{y}(t) + y(t) = 0.$$

a) Choose the state variable x, and write a state-space realization, $\dot{x}(t) = f(x(t))$ for some suitable function f; [3]

b) Discuss existence and unicity of solutions; [2]

c) Find all equilibria of the system; [2]

d) Linearize the system around each linearizable equilibrium and discuss the local phase-portrait; [3]

e) Find R so that the set $\{(y,\dot{y}) : y^2 + \dot{y}^2 \leq R^2\}$ is forward invariant for the system; [4]

f) Sketch the global phase portrait of the system. Do you expect periodic solutions to exist? Why? [4]

g) Show that the set $\{(y,\dot{y}) : y^2 + \dot{y}^2 = 1\}$ is an invariant set for the system. [2]

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2. Consider the following second order differential equation in the real variable y with external forcing input d:

$$\ddot{y}(t) = -\dot{y}(t) - \text{atan}(y(t)) + d(t).$$

a) Choose the state variable x, and write a state-space realization, $\dot{x}(t) = f(x(t), d(t))$ for some suitable function f; [3]

b) Consider first the unforced system, viz. $d(t) = 0$. Compute its equilibrium state and show that it is globally asymptotically stable. (*Hint: use the analog of mechanical energy as a candidate Lyapunov function.*) [7]

c) Assume now that $d(t)$ is specified by the following feedback law:

$$d(t) = \text{atan}(y(t)) + \dot{y}(t) + \frac{1}{1+t^2}.$$

i) Write equations describing the dynamics of the closed-loop system; [2]

ii) Argue that the new system can be seen as a linear system forced by an exogenous input signal and compute explicitly the solution $y(t)$ for an arbitrary initial condition $y(0), \dot{y}(0)$; [2]

iii) Show that the signal $\dot{y}(t)$, $t \geq 0$, is bounded; viz. there exists $M > 0$ such that $|\dot{y}(t)| \leq M$ for all $t \geq 0$; [2]

iv) Show that $d(t)$, as specified from the previous formula, is also a bounded signal; [2]

v) Remark that for $\dot{y}(0) = 0$, $y(t)$ is a diverging signal, and explain why this proves that our system is 0-GAS but not Input-to-State Stable. [2]

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3. Consider the following two-dimensional nonlinear control system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \sin(x_1)x_2 \\ -\sin(x_1)x_1 + u \end{bmatrix},$$

$$y = x_2.$$

a) Show that the system is passive from u to y (*Hint: use a quadratic storage function*); [4]

b) Consider the static output feedback law:

$$u(t) = -\text{atan}(y(t)).$$

Prove that the point $x = [0,0]'$ is a locally asymptotically stable equilibrium of the closed-loop system. [6]

c) Consider next a linear controller:

$$u(t) = -x_2(t) - x_1(t) + d(t).$$

where $d(t)$ is an exogenous input disturbance. Prove that the closed-loop system is Input-to-State Stable. (*Hint: use a quadratic ISS candidate Lyapunov function. Try to dominate non-definite terms*). [6]

d) Show that $[0,0]'$ is a Lyapunov stable equilibrium of the open-loop system ($u(t) = 0$) but not an asymptotically stable one. [4]

4. Consider the following three-dimensional nonlinear control system:

$$\begin{aligned} \dot{x}_1 &= \sin(x_2) - x_1 \\ \dot{x}_2 &= \sin(x_3) - x_2 \\ \dot{x}_3 &= u + \sin(x_1). \end{aligned}$$

with state $x = [x_1, x_2. x_3]' \in \mathbb{R}^3$ and input $u \in \mathbb{R}$.

a) Assume that $y = x_1$; compute the relative degree at $x = 0$ of the system. Specify if this is local or global relative degree. [3]

b) Find a feedback $u = v + k(x)$, where v denotes an auxiliary scalar control variable so that the resulting system is linear from input v to output y. [3]

c) Design a feedback law that globally asymptotically stabilizes the system at some equilibrium with output $y = x_1 = 0$. [3]

d) Consider next the output equation $y = x_3$ and compute the relative degree at 0 of the system. [2]

e) Highlight the internal dynamics and the zero-dynamics of the system corresponding to the output choice $y = x_3$. [4]

f) Show that the internal dynamics are ISS with respect to the exogenous signal y. Argue how this result can be used to design an output feedback controller capable of globally stabilizing the system around the equilibrium corresponding to $y = x_3 = 0$. [5]